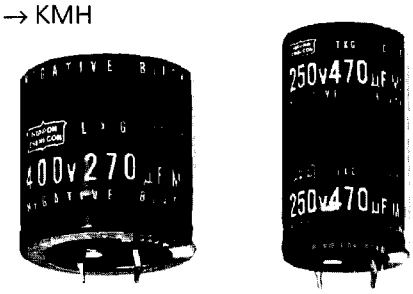
**LXG SERIES** • **105°C, 5,000 hours** LIFE: KMH → LX/LXA/LXG → TXG
SIZE: LX/LXA → LXG/TXG → KMH

**TXG SERIES** • **105°C, 10,000 hours**



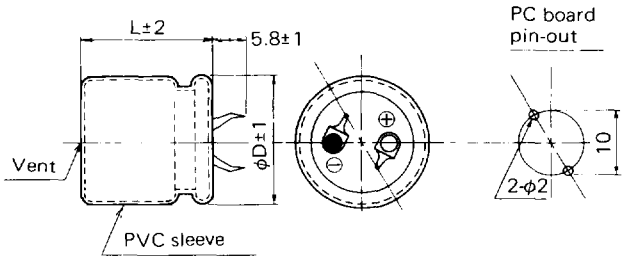

- Long life, specified with the rated ripple current applied.
- Non-solventproof type.
- LXG: 10 ~ 400V TXG: 10 ~ 250V

## CHARACTERISTICS

| Item | Characteristics |
|---|---|
| **Operating temperature range** | −40 ~ +105°C for 10 ~ 100V<br>−25 to +105°C for 200 ~ 400V (TXG is 250V or less.) |
| **Capacitance tolerance** | ±20% (M) (at 20°C, 120Hz) |
| **Leakage current** | 0.02CV (μA) or 3mA, whichever is smaller, after 5 minutes at 20°C. |
| **tan δ** | Rated voltage (V): 10 / 16 / 25 / 35 / 50 / 63~400<br>tan δ: 0.60 / 0.45 / 0.30 / 0.25 / 0.20 / 0.15 (at 20°C, 120Hz) |
| **Low temperature characteristics** | • Capacitance at the rated minimum temperature and 120Hz shall not less than 70% of the 20°C value.<br>• Impedance ratio at 120Hz shall not exceed the values given in the below table;<br>Rated voltage (V): 10~16 / 25 / 35 / 50~63 / 80~100 / 200~400<br>Z at −25°C/Z at +20°C: 4 / 3 / 3 / 2 / 2 / 4<br>Z at −40°C/Z at +20°C: 15 / 10 / 8 / 6 / 5 / – |
| **Impedance at 30KHz, 20°C** | The impedance shall not exceed the values in the table. |
| **Load life with ripple current applied** | After the capacitors are subjected to DC voltage with the rated ripple current applied at 105°C over the test time below, the following specifications shall be satisfied when the capacitors are restored to 20°C. The sum of DC voltage and peak AC voltage must not exceed the full rated voltage of the capacitor.<br>**LXG series; 5,000 hours**<br>**TXG series; 10,000 hours**<br>Capacitance change ≦ ±25% of the initial measured value<br>tan δ ≦ 250% of the initial specified value<br>Leakage current ≦ The initial specified value |
| **Shelf life** | The following specifications shall be satisfied when the capacitors are restored to 20°C after exposing them at 105°C for 500 hours without voltage applied.<br>Capacitance change ≦ ±20% of the initial measured value<br>tan δ ≦ 150% of the initial specified value<br>Leakage current ≦ The initial specified value |
| **Others** | Satisfies characteristic W of JIS C 5141. |

## DIMENSIONS

φD = φ22~φ35 mm (Type: VN) Unit (mm)




Note: • LXG – No epoxy type TXG – Epoxy type

## STANDARD RATINGS

LXG 10~400V

TXG 10~250V

| Case size øDxL(mm) | 10V | | 16V | | 25V | | 35V | | 50V | | 63V | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) |
| | µF | 105°C, 120Hz | µF | 105°C, 120Hz | µF | 105°C, 120Hz | µF | 105°C, 120Hz | µF | 105°C, 120Hz | µF | 105°C, 120Hz |
| 22x25 | 6,800 | 1.30 | 5,600 | 1.44 | 3,900 | 1.31 | 2,200 | 1.10 | 1,500 | 1.02 | 1,000 | 1.00 |
| 22x30 | 10,000 | 1.65 | 6,800 | 1.66 | 4,700 | 1.51 | 3,300 | 1.42 | 1,800 | 1.17 | 1,200 | 1.15 |
| 22x35 | 12,000 | 1.85 | 8,200 | 1.87 | 5,600 | 1.70 | 3,900 | 1.58 | 2,200 | 1.33 | 1,500 | 1.32 |
| 22x40 | 15,000 | 2.12 | 10,000 | 2.12 | 6,800 | 1.92 | 4,700 | 1.78 | 2,700 | 1.51 | 1,800 | 1.49 |
| 22x50 | 18,000 | 2.45 | 15,000 | 2.74 | 10,000 | 2.45 | 6,800 | 2.26 | 3,900 | 1.91 | 2,700 | 1.92 |
| 25.4x25 | 10,000 | 1.64 | 6,800 | 1.66 | 4,700 | 1.51 | 3,300 | 1.41 | 1,800 | 1.17 | 1,200 | 1.15 |
| 25.4x30 | 12,000 | 1.85 | 10,000 | 2.07 | 6,800 | 1.87 | 3,900 | 1.58 | 2,700 | 1.47 | 1,800 | 1.45 |
| 25.4x35 | 15,000 | 2.16 | 12,000 | 2.37 | 8,200 | 2.14 | 5,600 | 1.98 | 3,300 | 1.70 | 2,200 | 1.67 |
| 25.4x40 | 18,000 | 2.43 | 15,000 | 2.71 | 10,000 | 2.43 | 6,800 | 2.24 | 3,900 | 1.89 | 2,700 | 1.90 |
| 25.4x50 | 27,000 | 3.11 | 18,000 | 3.11 | 12,000 | 2.78 | 8,200 | 2.57 | 5,600 | 2.38 | 3,300 | 2.20 |
| 30x25 | 12,000 | 1.89 | 10,000 | 2.11 | 6,800 | 1.90 | 4,700 | 1.77 | 2,700 | 1.50 | 1,800 | 1.48 |
| 30x30 | 18,000 | 2.37 | 12,000 | 2.37 | 8,200 | 2.15 | 5,600 | 1.98 | 3,300 | 1.70 | 2,200 | 1.68 |
| 30x35 | 22,000 | 2.73 | 18,000 | 3.02 | 12,000 | 2.70 | 8,200 | 2.50 | 4,700 | 2.11 | 2,700 | 1.93 |
| 30x40 | 27,000 | 3.13 | 22,000 | 3.46 | 15,000 | 3.13 | 10,000 | 2.86 | 5,600 | 2.39 | 3,900 | 2.41 |
| 30x50 | 39,000 | 3.99 | 27,000 | 4.07 | 18,000 | 3.64 | 12,000 | 3.32 | 6,800 | 2.79 | 4,700 | 2.80 |
| 35x25 | 18,000 | 2.42 | 12,000 | 2.42 | 8,200 | 2.19 | 5,600 | 2.03 | 3,300 | 1.74 | 2,200 | 1.71 |
| 35x30 | 22,000 | 2.79 | 18,000 | 3.09 | 12,000 | 2.76 | 8,200 | 2.55 | 4,700 | 2.16 | 3,300 | 2.18 |
| 35x35 | 33,000 | 3.49 | 22,000 | 3.49 | 15,000 | 3.16 | 10,000 | 2.88 | 5,600 | 2.41 | 3,900 | 2.43 |
| 35x40 | 39,000 | 3.96 | 27,000 | 4.04 | 18,000 | 3.61 | 12,000 | 3.30 | 6,800 | 2.78 | 4,700 | 2.78 |
| 35x50 | 47,000 | 4.62 | 39,000 | 5.16 | 27,000 | 4.70 | 18,000 | 4.29 | 10,000 | 3.57 | 6,800 | 3.55 |

| Case size øDxL(mm) | 80V | | 100V | | 200V | | 250V | | 350V LXG | | 400V LXG | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) | Capacitance | Ripple current (A rms) |
| | µF | 105°C, 120Hz | µF | 105°C, 120Hz | µF | 105°C, 120Hz | µF | 105°C, 120Hz | µF | 105°C, 120Hz | µF | 105°C, 120Hz |
| 22x25 | 680 | 0.97 | 390 | 0.78 | 180 | 0.63 | 150 | 0.57 | 68 | 0.39 | 56 | 0.39 |
| 22x30 | 820 | 1.12 | 560 | 0.99 | 270 | 0.81 | 180 | 0.66 | 82 | 0.49 | 68 | 0.44 |
| 22x35 | 1,000 | 1.27 | 680 | 1.12 | 330 | 0.92 | 220 | 0.75 | 100 | 0.55 | 82 | 0.50 |
| 22x40 | 1,200 | 1.42 | 820 | 1.26 | 390 | 1.02 | 270 | 0.85 | 120 | 0.63 | 120 | 0.57 |
| 22x50 | 1,800 | 1.84 | 1,200 | 1.60 | 470 | 1.17 | 390 | 1.08 | 180 | 0.73 | 150 | 0.67 |
| 25.4x25 | 1,000 | 1.23 | 560 | 0.98 | 270 | 0.81 | 180 | 0.66 | 100 | 0.49 | 68 | 0.40 |
| 25.4x30 | 1,200 | 1.39 | 820 | 1.23 | 330 | 0.92 | 270 | 0.83 | 120 | 0.55 | 100 | 0.50 |
| 25.4x35 | 1,500 | 1.62 | 1,000 | 1.41 | 470 | 1.15 | 330 | 0.96 | 150 | 0.60 | 120 | 0.58 |
| 25.4x40 | 1,800 | 1.82 | 1,200 | 1.59 | 560 | 1.28 | 390 | 1.07 | 180 | 0.72 | 150 | 0.66 |
| 25.4x50 | 2,200 | 2.11 | 1,500 | 1.86 | 680 | 1.48 | 470 | 1.22 | 220 | 0.93 | 220 | 0.84 |
| 30x25 | 1,200 | 1.41 | 820 | 1.25 | 330 | 0.94 | 270 | 0.85 | 120 | 0.56 | 100 | 0.51 |
| 30x30 | 1,800 | 1.78 | 1,000 | 1.42 | 470 | 1.15 | 330 | 0.96 | 180 | 0.71 | 150 | 0.65 |
| 30x35 | 2,200 | 2.05 | 1,200 | 1.61 | 560 | 1.30 | 470 | 1.19 | 220 | 0.82 | 180 | 0.74 |
| 30x40 | 2,700 | 2.35 | 1,500 | 1.87 | 680 | 1.49 | 560 | 1.35 | 270 | 0.94 | 220 | 0.84 |
| 30x50 | 3,300 | 2.75 | 2,200 | 2.40 | 1,000 | 1.91 | 680 | 1.58 | 330 | 1.19 | 270 | 1.10 |
| 35x25 | 1,800 | 1.82 | 1,000 | 1.45 | 470 | 1.22 | 330 | 1.02 | 150 | 0.69 | 120 | 0.61 |
| 35x30 | 2,200 | 2.09 | 1,500 | 1.85 | 680 | 1.47 | 470 | 1.22 | 220 | 0.83 | 180 | 0.83 |
| 35x35 | 2,700 | 2.37 | 1,800 | 2.07 | 820 | 1.65 | 560 | 1.36 | 270 | 1.04 | 220 | 0.94 |
| 35x40 | 3,300 | 2.73 | 2,200 | 2.39 | 1,000 | 1.90 | 680 | 1.57 | 330 | 1.19 | 270 | 1.09 |
| 35x50 | 4,700 | 3.46 | 2,700 | 2.81 | 1,200 | 2.21 | 1,000 | 2.02 | 470 | 1.38 | 390 | 1.26 |

**Note**: These specifications are common for both LXG and TXG, except 350 and 400V are only applied to LXG.

## MAX. IMPEDANCE (mΩ) at 20°C, 30KHz

| Series / Case size øDxL (mm) \ V | LXG/TXG | | | LXG |
|---|---|---|---|---|
| | 10 ~ 63 | 80 ~ 100 | 200 ~ 250 | 350 ~ 400 |
| 22x25 | 120 | 150 | 640 | 800 |
| 22x30 | 100 | 120 | 560 | 700 |
| 22x35 | 80 | 95 | 450 | 560 |
| 22x40 | 70 | 80 | 400 | 500 |
| 22x50 | 50 | 60 | 320 | 390 |
| 25.4x25 | 90 | 110 | 530 | 660 |
| 25.4x30 | 70 | 85 | 420 | 520 |
| 25.4x35 | 60 | 70 | 360 | 440 |
| 25.4x40 | 50 | 60 | 310 | 380 |
| 25.4x50 | 40 | 45 | 240 | 300 |
| 30x25 | 70 | 80 | 400 | 500 |
| 30x30 | 50 | 60 | 320 | 400 |
| 30x35 | 40 | 50 | 260 | 320 |
| 30x40 | 35 | 40 | 240 | 290 |
| 30x50 | 25 | 30 | 180 | 220 |
| 35x25 | 65 | 70 | 340 | 420 |
| 35x30 | 45 | 50 | 260 | 320 |
| 35x35 | 38 | 40 | 210 | 260 |
| 35x40 | 30 | 30 | 180 | 220 |
| 35x50 | 23 | 25 | 140 | 170 |

## RIPPLE CURRENT MULTIPLIERS LXG/TXG

### Frequency Multiplier:

| Frequency / Rated voltage | 50Hz | 120Hz | 300Hz | 1KHz | 10KHz | 50KHz |
|---|---|---|---|---|---|---|
| 10~50V | 0.95 | 1.00 | 1.03 | 1.05 | 1.08 | 1.08 |
| 63~100V | 0.92 | 1.00 | 1.07 | 1.13 | 1.19 | 1.20 |
| 200~250V | 0.81 | 1.00 | 1.17 | 1.32 | 1.45 | 1.50 |
| 350~400V (LXG) | 0.77 | 1.00 | 1.16 | 1.30 | 1.41 | 1.43 |

### Temperature Multiplier:

| Ambient Temperature (°C) | 85 | 105 |
|---|---|---|
| Factor | 1.73 | 1.00 |

Note: When long life performance is required in actual use, the rms ripple current has to be reduced.



Specifications subject to change without notice.